

EE 473: Analysis of Epileptic Seizures using Wavelet Transforms

Project Report

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1 Project Member Responsibilities

Aziz Bakirci was responsible for research and implementation & visualization of the Continuous Wavelet Transform, including the comparison with the Fourier Transform. He also implemented Fourier-based classification and comparison with the Wavelet-based approach.

Boran Aybak Kilic conducted research Discrete Wavelet Transform and implementation using Mallat’s algorithm. He also created the pipeline including denoising, feature extraction and classification methods for processing and classifying EEG signals on Python alongside functions for visualizations and evaluation.

2 Problem Statement

Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures, affecting millions worldwide. The primary diagnostic tool, the Electroencephalogram (EEG), records the brain’s electrical activity. However, visual inspection of these signals is time-consuming and error-prone due to the transient nature of ictal (seizure) events.

The core technical challenge addressed in this project is the non-stationary nature of EEG signals. Traditional Fourier Transform (FT) methods provide frequency information but lose time localization, making them unsuitable for detecting brief epileptic spikes. **This project aims to develop an automated classification system using Wavelet Transforms, which function as a “mathematical microscope” to provide the necessary multi-resolution analysis for reliable seizure detection.**

3 Introduction

Epilepsy is a chronic neurological disorder affecting over 50 million people globally, characterized by recurrent, unprovoked seizures that result from sudden, abnormal electrical discharges in the brain [1, 2]. Electroencephalography (EEG) remains the primary non-invasive tool for monitoring these electrical potentials, which reflect the summation of post-synaptic activities from millions of neurons [3, 4]. The history of EEG analysis dates back to the work of Hans Berger, who first recorded alpha wave activity in the 1870s [5, 6]. For decades, clinical diagnosis relied on manual visual inspection by neurophysiologists; however, this manual process is labor-intensive, subjective, and prone to human error, as brief ictal events can be missed within long-term recordings [7, 8, 9]. To overcome the limitations of qualitative analysis, researchers turned to quantitative spectral techniques, primarily the **Fourier Transform (FT)**. The FT decomposes a time-domain signal into its constituent sinusoidal frequency components [10, 11]. While powerful for stationary signals, legacy frequency analysis is insufficient for EEG data because brain signals are inherently non-stationary, non-linear, and transient [1, 12]. The traditional FT provides global frequency information but lacks temporal localization; it can identify what frequencies are present, but not when they occurred [13, 14, 15]. The **Short-Time Fourier Transform (STFT)** attempted to resolve this by applying the FT to successive, overlapping windows of the signal [16, 14]. However, the STFT is fundamentally constrained by the **Heisenberg Uncertainty Principle** (or Gabor Limit), which states that one cannot achieve perfect resolution in both time and frequency domains simultaneously [13, 14]. Because the STFT uses a fixed window size, it forces a uniform resolution

trade-off across all frequencies: a narrow window provides good time resolution but poor frequency resolution, while a wide window does the reverse [13].

This project utilizes the **Wavelet Transform (WT)** to transcend these constraints. Often described as a "mathematical microscope," the WT provides a multi-resolution analysis (MRA) by employing short, oscillating functions called mother wavelets [4, 17]. Unlike the fixed windows of the STFT, wavelets utilize adaptive windowing: they provide high time resolution for high-frequency events, such as seizure spikes, and high frequency resolution for low-frequency underlying rhythms [13, 4]. The theoretical foundation of MRA, initiated by **Stephane Mallat**, treats a signal as a sum of approximations and details across nested vector spaces [18, 9]. For computational efficiency, the **Discrete Wavelet Transform (DWT)** is implemented via **Mallat's Algorithm** (the Fast Wavelet Transform) [19]. This algorithm uses a recursive filter bank consisting of low-pass scaling filters (Father wavelets) and high-pass wavelet filters (Mother wavelets) [8]. Through a process of downsampling by two at each level, the signal is split into approximation (low-frequency) and detail (high-frequency) coefficients [9]. These levels correspond directly to biological brain waves: Delta, Theta, Alpha, Beta, and Gamma [7]. In this project, we have developed an automated detection pipeline using the "**Epileptic Seizure Recognition**" dataset. Our work involves signal preprocessing through normalization and denoising using the VisuShrink algorithm with soft thresholding to remove Gaussian noise while preserving seizure spikes [20]. We implemented a 5-level DWT using various wavelet families, such as Daubechies (db4, db2) and Symlets, to extract discriminative features, including relative Energy, Shannon Entropy, and Standard Deviation from each sub-band [21]. These features were fed into Support Vector Machine (SVM) and XGBoost classifiers. Our results confirm that wavelet-based features provide significantly higher accuracy than traditional Fourier-based methods, achieving near-perfect classification in distinguishing seizure activity from healthy brain states [22].

4 Methodology

4.1 The Need For the Wavelet Transform

4.1.1 Continuous Fourier Transform (CFT)

In general, an integral transform on $f \in L^2$, which are the functions that are square integrable, i.e $\int |f| < \infty$, can be defined with a mathematical operation called *inner product*. It is defined as follows with $f(t)$ as *signal* and $g(t)$ as *analysis function* or *kernel*:

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(t) \cdot g^*(t) dt \quad (1)$$

As can be seen, if we define our *analysis function*, $g(t)$ as $e^{j\omega t}$, then the inner-product comes as follows:

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} dt \quad (2)$$

which is our legacy Fourier Transform. The greatest property of the Fourier Transform is also its weakness; it captures the frequency content with infinite resolution, as great as it is for analyzing the static function's frequency content, due to Heisenberg's Indeterminacy, or more popularly as Heisenberg's Uncertainty, it completely lacks any

information regarding temporal information. The more formal explanation from the formal definitions of Heisenberg's work regarding time spread(uncertainty) and frequency spread(uncertainty):

$$\Delta t^2 = \frac{\int (t - \bar{t}) |g(t)|^2 dt}{\int |g(t)|^2 dt}, \quad \text{where} \quad \bar{t} = \frac{\int t |g(t)|^2 dt}{\int |g(t)|^2 dt} \quad (3)$$

and similarly;

$$\Delta \omega^2 = \frac{\int (\omega - \bar{\omega}) |G(\omega)|^2 d\omega}{\int |G(\omega)|^2 d\omega}, \quad \text{where} \quad \bar{\omega} = \frac{\int \omega |G(\omega)|^2 d\omega}{\int |G(\omega)|^2 d\omega} \quad (4)$$

notice that we have used $g(t)$ and $G(\omega)$ for the expressions. The reason we have used our kernels is that any transform's resolution capabilities are restricted by those of their kernels. Now, if we were to give our Fourier Transform kernel of $e^{j\omega t}$, then we get $\bar{\omega} = \omega$ and hence $\Delta \omega = 0$. As can be seen, the frequency content is captured with perfect resolution; however, when we look at the time uncertainty, it is ∞ . Therefore, we can see that the ability of the Fourier Transform lies purely in capturing the frequency content at the cost of temporal information. In order to overcome this problem, the Short-Time Fourier Transform can be implemented.

4.1.2 Short Time Fourier Transform (STFT)

In order to capture temporal information at the expense of reduced frequency resolution, the *Short Time Fourier Transform* (STFT), also referred to as the *Windowed Fourier Transform*, was developed. It is defined by imposing a windowing function on the complex exponential kernel:

$$F(t_0, \omega) = \int_{-\infty}^{\infty} f(t) g^*(t - t_0) e^{-j\omega t} dt. \quad (5)$$

In this expression, $g(t)$ denotes the windowing function, which is typically chosen to be either band-limited or rapidly decaying in time. The presence of the window function localizes the analysis around the time instant t_0 , allowing the transform to capture both temporal and spectral characteristics of the signal. Consequently, $F(t_0, \omega)$ provides information about how the frequency content of the signal evolves over time.

However, it is important to emphasize that the STFT does not constitute a universally superior analysis method, as it remains subject to the time–frequency uncertainty principle. The resolution of the transform is fundamentally limited by the windowing function: improving time resolution by narrowing the window inevitably results in a loss of frequency resolution, while improving frequency resolution by widening the window leads to reduced temporal localization. This trade-off is intrinsic and cannot be eliminated.

An important characteristic of the STFT is that the time–frequency resolution is *uniform* over the entire analysis domain. Since the same windowing function $g(t)$ is used for all time shifts t_0 and all modulation frequencies ω , the effective time spread Δt and frequency spread $\Delta \omega$ of the analysis kernel remain constant. As a result, each STFT coefficient corresponds to a fixed-resolution region in the time–frequency plane.

This fixed resolution is often illustrated using a *Heisenberg box*, which represents the minimum region of joint time–frequency localization associated with the analysis kernel. In the case of the STFT, this Heisenberg box has a constant rectangular shape and area,

determined solely by the window function $g(t)$ and constrained by the time–frequency uncertainty principle. Consequently, the STFT tiles the time–frequency plane using identical resolution cells, regardless of frequency or time location.

While this uniform tiling provides a consistent interpretation of time–frequency content, it also implies that the STFT cannot adapt its resolution to different signal components. High-frequency components, which may benefit from finer temporal localization, and low-frequency components, which may require finer frequency resolution, are analyzed using the same fixed Heisenberg box. This lack of adaptability motivates the use of alternative representations, such as wavelet transforms, in which the resolution cells vary with scale.

4.1.3 Continuous Wavelet Transform (CWT)

The CWT is used primarily for time–frequency visualization (Scalograms). It correlates the signal $x(t)$ with a mother wavelet $\psi(t)$ that is scaled by parameter a and translated by parameter b . The mathematical definition is given by:

$$W(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t - b}{a} \right) dt \quad (6)$$

Where:

- a controls the dilation (scale), inversely related to frequency ($f \propto 1/a$).
- b controls the translation (time localization).
- ψ^* denotes the complex conjugate of the mother wavelet.

By varying the scale parameter a , the wavelet kernel achieves different time–frequency resolutions: small values of a correspond to wavelets that are well localized in time but spread in frequency, whereas large values of a yield wavelets that are well localized in frequency but extended in time. Importantly, although the shape of the Heisenberg box changes with scale, its area remains bounded by the uncertainty principle, ensuring a constant minimum joint time–frequency localization.

This adaptive resolution property allows wavelet transforms to provide a multiresolution analysis of signals, making them particularly suitable for analyzing signals that exhibit nonstationary or multiscale behavior. The difference caused by varying resolution versus constant resolution in Heisenberg Boxes which occurs with STFT is illustrated as follows[23]:

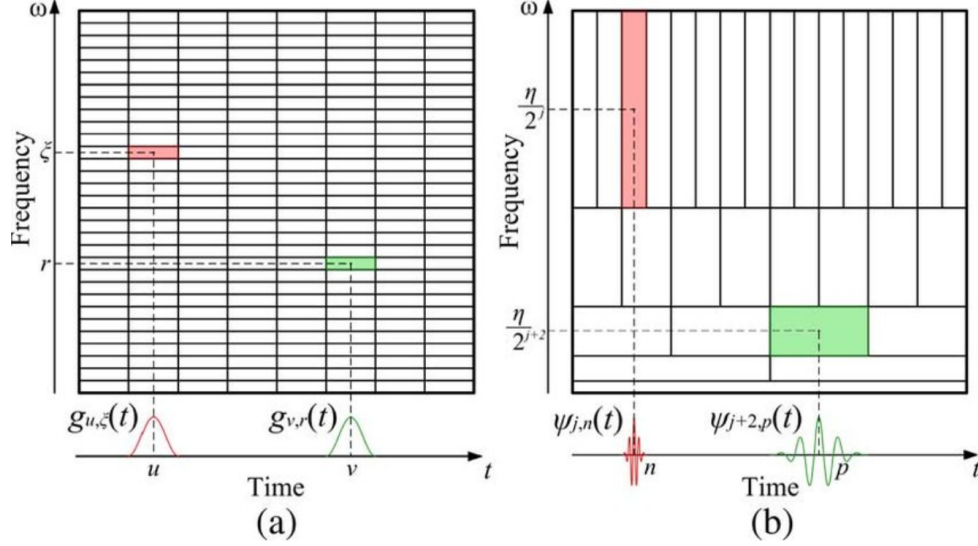


Figure 1: Heisenberg Boxes for STFT (a) and Wavelet Transform (b)

4.2 Discrete Wavelet Transform (DWT)

4.2.1 Scaling (Father) and Wavelet Functions (Mother Wavelets)

Wavelet analysis is built upon two fundamental functions: the **scaling function** $\varphi(t)$ (often called the "Father Wavelet") and the **wavelet function** $\psi(t)$ (the "Mother Wavelet").

4.2.1.1 The Father Wavelet: Scaling Function

The scaling function $\varphi(t)$ is essential for covering the low-frequency approximation of a signal. It defines the basic resolution of the analysis. A family of scaling functions is the set of basis functions composed of all integer translations and binary scalings of the real, square-integrable father scaling function $\varphi(x)$ —that is, the set of all scaled and translated functions $\{\varphi_{j,k}(x) | j, k \in \mathbb{Z}\}$ where

$$\varphi_{j,k}(t) = 2^{j/2} \varphi(2^j t - k) \quad (7)$$

where j determines the scale (width) and k determines the position. These functions span a subspace $V_j \subset L^2(\mathbb{R})$, the set of square integrable, 1-D functions. Note that scaling function is orthogonal to its integer translates.

4.2.1.2 The Mother Wavelet: Wavelet Function

The wavelet function $\psi(t)$ encodes the details or differences between approximations at adjacent scales. Like the scaling function, it generates a family:

$$\psi_{j,k}(t) = 2^{j/2} \psi(2^j t - k) \quad (8)$$

The wavelets span the subspace W_j , which is the orthogonal complement of V_j in V_{j+1} .

$$\varphi_{j,k}(x) \in V_j, \psi_{j,k}(x) \in W_j \Rightarrow V_j \oplus W_j = V_{j+1} \quad (9)$$

where $\oplus$ denotes the union of function spaces.

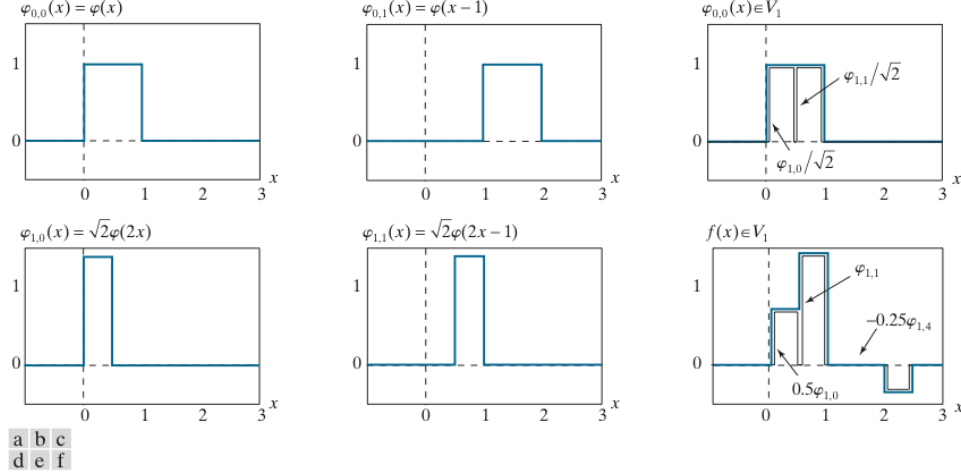


FIGURE 7.19 The Haar scaling function.

Figure 2: Haar Scaling Function and figures showing how a scaling function can be expressed as a sum of finer scale father wavelets.[24]

4.2.2 Multiresolution Analysis (MRA) and Vector Spaces

The interaction between these functions is governed by the axioms of Multiresolution Analysis (MRA).

4.2.2.1 The Four MRA Properties

For the scaling function $\varphi(x)$ to generate a valid Multiresolution Analysis, it must satisfy the following four conditions:

1. **Orthogonality:** The scaling function is orthogonal to its integer translates.
2. **Nesting:** The subspaces spanned by the scaling function at low scales are nested within those at higher scales:

$$V_{-\infty} \subset \cdots \subset V_{-1} \subset V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_{\infty} \quad (10)$$

3. **Zero Intersection:** The only function that is common to all V_j is the zero function:

$$\bigcap_{j \in \mathbb{Z}} V_j = \{0\} \quad (11)$$

4. **Completeness:** Any function can be represented with arbitrary precision (the union of the subspaces is dense in $L^2(\mathbb{R})$):

$$V_{\infty} = \bigcup_{j \in \mathbb{Z}} V_j = L^2(\mathbb{R}) \quad (12)$$

4.2.2.2 Orthogonal Vector Spaces

The beauty of wavelet theory lies in the decomposition of these spaces. The space V_{j+1} (finer resolution) consists of the coarse approximation V_j plus the added details W_j .

$$V_{j+1} = V_j \oplus W_j \quad (13)$$

Here, $\oplus$ denotes the orthogonal sum. This implies two crucial orthogonality conditions:

- The spaces are mutually orthogonal: $V_j \perp W_j$.
- The basis functions within W_j are orthogonal to those in V_j :

$$\langle \varphi_{j,k}(t), \psi_{j,m}(t) \rangle = 0 \quad \text{for all } k, m \quad (14)$$

By iterating Equation 13, we can express the space of all measurable functions $L^2(\mathbb{R})$ as the sum of a very coarse approximation (V_{j_0}) and all subsequent detail spaces:

$$L^2(\mathbb{R}) = V_{j_0} \oplus W_{j_0} \oplus W_{j_0+1} \oplus \dots \quad (15)$$

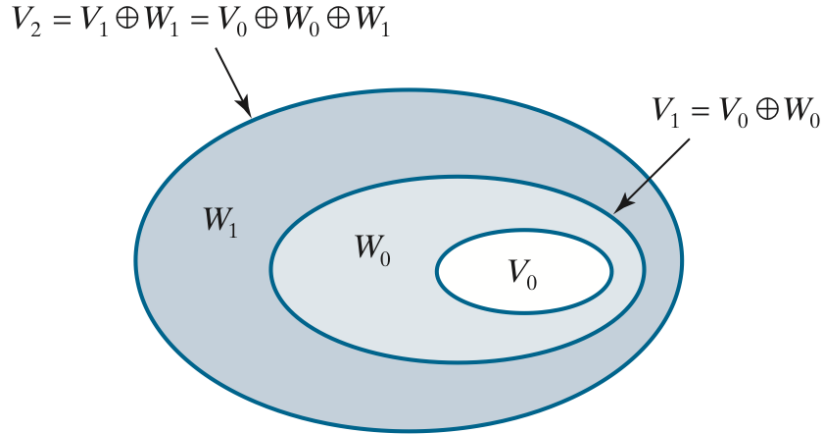


Figure 3: Visual representation of nested subspaces and orthogonal complements.[24]

4.2.3 The Dilation Equations and Filter Coefficients

Since $V_0 \subset V_1$, both the scaling function $\varphi(t)$ and the wavelet $\psi(t)$ (which reside in V_0 and W_0 respectively) can be represented as linear combinations of the basis functions of V_1 (which are $\varphi(2t)$). This relationship is defined by the **Dilation Equations**:

$$\varphi(t) = \sum_n h_\varphi[n] \sqrt{2} \varphi(2t - n) \quad (16)$$

$$\psi(t) = \sum_n h_\psi[n] \sqrt{2} \varphi(2t - n) \quad (17)$$

Here, $h_\varphi[n]$ are the scaling filter coefficients (low-pass) and $h_\psi[n]$ are the wavelet filter coefficients (high-pass). These coefficients determine the specific wavelet family (e.g., Haar, Daubechies). Scaling filter coefficients can be determined by the inner product

$$h_\varphi[n] = \langle \varphi(x), \sqrt{2} \varphi(2x - n) \rangle. \quad (18)$$

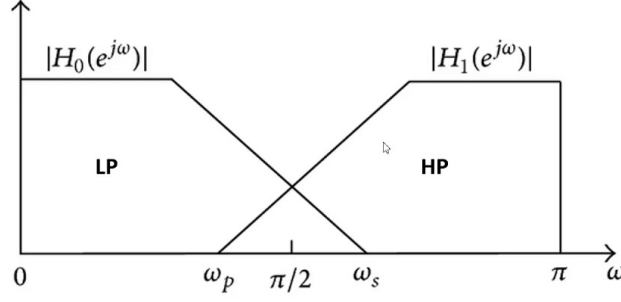


Figure 4: Quadrature mirror filters spanning the low and high frequency components.

4.2.4 Relationship Between Filters: The Quadrature Mirror Filter

For the wavelet basis to be orthogonal, the filter coefficients h_φ and h_ψ are not independent. They are related by the Quadrature Mirror Filter (QMF) relationship:

$$h_\psi[n] = (-1)^n h_\varphi[1 - n] \quad (19)$$

This relationship will prove very useful in Fast Wavelet Transform 4.2.6 since it allows us to determine wavelet and scaling coefficients without explicitly calculating both. Their Fourier transforms are also related by

$$H_\psi(\omega) = -e^{-j\omega} H_\varphi(\pi - \omega). \quad (20)$$

Above equation shows that scaling and wavelet function functions form a complementary basis in the frequency domain as well as acting as a low-pass and high-pass filter pair. For real-valued filters, $H(\pi - \omega) = H^*(\pi + \omega)$, ensures the two filters are orthogonal to each other, allowing for perfect reconstruction of the signal.

Proof of the Filter Relationship

We require the scaling function $\varphi(t)$ and wavelet $\psi(t)$ to be orthogonal: $\langle \varphi(t), \psi(t) \rangle = 0$. Substituting the dilation equations (16) and (17):

$$\begin{aligned} \langle \varphi(t), \psi(t) \rangle &= \left\langle \sum_n h_\varphi[n] \sqrt{2} \varphi(2t - n), \sum_k h_\psi[k] \sqrt{2} \varphi(2t - k) \right\rangle \\ &= \sum_n \sum_k h_\varphi[n] h_\psi[k] \cdot 2 \underbrace{\langle \varphi(2t - n), \varphi(2t - k) \rangle}_{\frac{1}{2} \delta[n-k]} \end{aligned}$$

The inner product of the shifted $\varphi(2t)$ functions is $\frac{1}{2} \delta[n - k]$ due to the orthonormality of the basis in V_1 . Thus, the condition simplifies to:

$$\sum_n h_\varphi[n] h_\psi[n] = 0 \quad (21)$$

This states that the filter vectors must be orthogonal. The specific choice $h_\psi[n] = (-1)^n h_\varphi[1-n]$ satisfies this condition. Let us verify by substitution into the sum (assuming real coefficients):

$$\sum_n h_\varphi[n](-1)^n h_\varphi[1-n] = 0$$

This sum is identically zero due to the symmetry and alternating signs, satisfying the orthogonality requirement. $\square$

4.2.5 The Discrete Wavelet Transform (DWT)

When applying this theory to digital signals, we deal with discrete sequences rather than continuous functions. Consider an input signal $x[n]$ of length N .

4.2.5.1 Discrete Constraints

- The signal length N is typically required to be a power of 2, i.e., $N = 2^J$.
- If the sequence length is not a power of 2, it is standard practice to **pad** the sequence (usually with zeros or by extending the boundary values) to the closest power of 2.

4.2.5.2 Approximation and Detail Coefficients

The DWT decomposes the signal $x[n]$ into approximation coefficients T_φ and detail coefficients T_ψ . Discrete wavelet expansion is given by

$$x[n] = \frac{1}{\sqrt{N}} \left[T_\varphi(0,0)\varphi[n] + \sum_{j=0}^{J-1} \sum_{k=0}^{2^j-1} T_\psi(j,k)\psi_{j,k}[n] \right] \quad (22)$$

At a specific scale j and translation k , these are given by the discrete inner products

$$T_\varphi(j,k) = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n]\varphi_{j,k}[n] \quad (23)$$

$$T_\psi(j,k) = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n]\psi_{j,k}[n] \quad (24)$$

The coefficient $T_\varphi(j,k)$ represents the coarse information (low frequencies) of $x[n]$ at scale j , while $T_\psi(j,k)$ captures the detailed information (high frequencies). In the next phase of our study, we will explore the *Fast Wavelet Transform*, which computes these values efficiently using recursive filter banks.

4.2.6 The Fast Wavelet Transform (FWT)

Calculating the wavelet coefficients $T_\varphi(j,k)$ and $T_\psi(j,k)$ directly via the inner products (Eqs. 23 and 24) for every scale j and translation k is computationally inefficient. Such an approach fails to exploit the redundancy inherent in the Multiresolution Analysis (MRA) framework.

The fundamental insight of the Fast Wavelet Transform (FWT) is that we do not need to calculate coefficients by revisiting the original signal $x[n]$ for every scale. Because the vector spaces are nested ($V_j \subset V_{j+1}$) and complementary ($V_{j+1} = V_j \oplus W_j$), the

information contained in the approximation coefficients at a finer scale $j + 1$ is sufficient to compute both the approximation and detail coefficients at the coarser scale j . This allows for an efficient, iterative algorithm with linear time complexity $O(N)$.

4.2.6.1 Derivation of the Recursive Analysis Equations

We seek a relationship that expresses the coefficients at scale j in terms of the coefficients at scale $j + 1$. Consider the definition of the approximation coefficient:

$$T_\varphi(j, k) = \langle x[n], \varphi_{j,k}[n] \rangle \quad (25)$$

Recall the relationship between basis functions at adjacent scales derived from the dilation equation:

$$\varphi_{j,k}(t) = \sum_m h_\varphi(m - 2k) \varphi_{j+1,m}(t) \quad (26)$$

Substituting this expansion into the inner product yields:

$$\begin{aligned} T_\varphi(j, k) &= \left\langle x[n], \sum_m h_\varphi(m - 2k) \varphi_{j+1,m}[n] \right\rangle \\ &= \sum_m h_\varphi(m - 2k) \langle x[n], \varphi_{j+1,m}[n] \rangle \end{aligned} \quad (27)$$

Observing that the inner product term $\langle x, \varphi_{j+1,m} \rangle$ is simply the approximation coefficient at the finer scale $T_\varphi(j + 1, m)$, we arrive at the recursive analysis equation:

$$T_\varphi(j, k) = \sum_n h_\varphi(n - 2k) T_\varphi(j + 1, n) \quad (28)$$

By applying the same logic to the wavelet function $\psi_{j,k}$, utilizing the wavelet dilation equation and the definition $T_\psi(j, k) = \langle x, \psi_{j,k} \rangle$, we derive the formula for the detail coefficients:

$$T_\psi(j, k) = \sum_n h_\psi(n - 2k) T_\varphi(j + 1, n) \quad (29)$$

Note: In these equations, n serves as the summation index over the finer scale coefficients.

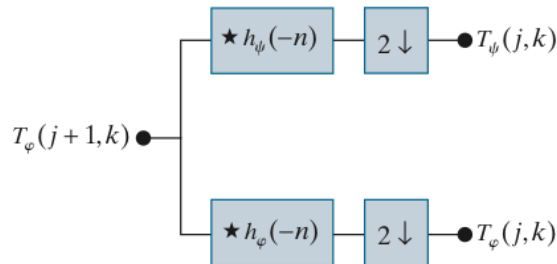


Figure 5: One stage of the Fast Wavelet Transform analysis filter bank.[24]

4.2.6.2 Filter Bank Implementation

Equations (28) and (29) reveal that the coefficients at scale j are computed by convolving the coefficients at scale $j + 1$ with the time-reversed scaling filter $h_\varphi(-n)$ and wavelet filter $h_\psi(-n)$, respectively, followed by downsampling by a factor of 2 (subsampling at indices $2k$).

$$T_\varphi(j, k) = \left(T_\varphi(j + 1, n) \star h_\varphi(-n) \right)_{\downarrow 2} \quad (30)$$

$$T_\psi(j, k) = \left(T_\varphi(j + 1, n) \star h_\psi(-n) \right)_{\downarrow 2} \quad (31)$$

where $\downarrow 2$ represents downsampling by 2

$$y_{\downarrow 2}[n] = y[2n], \quad \text{for } n = 0, 1, 2, \dots \quad (32)$$

This structure naturally forms a **recursive filter bank**. The algorithm begins with the original signal $x[n]$ (assumed to be the coefficients at the finest scale J) and iteratively "peels off" layers of detail.

The process repeats on the approximation coefficients $T_\varphi(j, k)$, decomposing them further into coarser approximations and details. The detail coefficients $T_\psi(j, k)$ are stored and not decomposed further. This cascade continues until the desired coarse scale is reached.

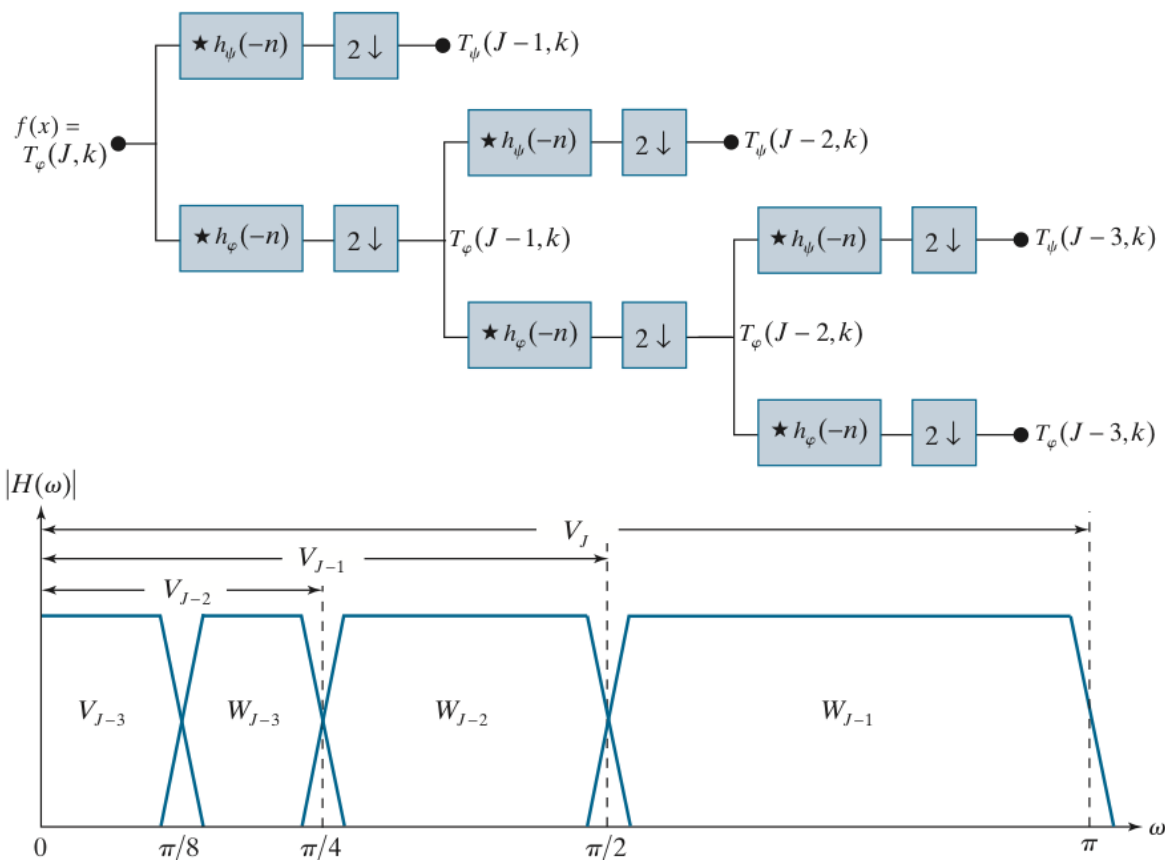


Figure 6: A three-stage FWT analysis filter bank. Note how the approximation coefficients are iteratively decomposed, creating a logarithmic division of the frequency spectrum.[24]

Figure 6 illustrates a 3-stage decomposition. The frequency domain interpretation shows that the FWT effectively splits the bandwidth of the signal, with the approximation T_φ representing the low-frequency band (e.g., 0 to $\pi/8$) and the various T_ψ levels representing octave-band high-frequency details.

4.3 Wavelet Families

We selected specific wavelet families based on signal morphology:

- **Morlet Wavelet:** Used for CWT due to its complex sinusoidal shape modulated by a Gaussian, providing excellent time-frequency localization.

$$\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-t^2/2} \quad (33)$$

- **Daubechies (db4):** Used for DWT due to its compact support and orthogonality, effectively capturing the transient spikes of epileptic seizures. It is obtained by solving the non-linear equations for filter coefficients while imposing minimum-phase.
- **Symlet (sym):** Used for DWT due to the fact that it is 'the least asymmetrical' solution of the filter coefficients equations. It is obtained by choosing the solution for the filter equation as close to linear-phase as possible. Its nearly symmetric property helps reducing the wrap-around error of reconstruction.
- **Coiflet(coif):** Used for DWT for its property of imposing vanishing moments on both scaling function and mother wavelet. Generally it is more desired for solving differential equations, which was the reason it was developed at the first place.
- **Biorthogonal (bior):** Used for DWT since according to Daubechies Theorem [25], it is not possible for a compact, real wavelet to be orthogonal and symmetrical at the same time. While Symlet tries to be as symmetrical as possible while being orthogonal, Biorthogonal Wavelet tries the opposite, which is being symmetrical by relaxing the orthogonality. The disadvantage comes within is that the analysis and synthesis kernels have completely different filter coefficients that have to be solved for each one of them.

4.3.1 Derivation of Wavelet Filter Coefficients

The wavelet filter coefficients $h[k]$ for the lowpass decomposition filter are derived by solving a system of nonlinear algebraic equations arising from three fundamental constraints. First, the normalization constraint

$$\sum_k h[k] = \sqrt{2} \quad (34)$$

ensures proper scaling of the decomposition. Second, the orthonormality constraints

$$\sum_k h[k]^2 = 1 \quad (35)$$

$$\sum_k h[k]h[k-2m] = \delta(m) \quad (36)$$

guarantee perfect reconstruction without aliasing. Third, the vanishing moments constraints

$$\sum_k k^m h[k] = 0 \quad (37)$$

ensure that the corresponding wavelet can represent smooth functions efficiently by annihilating polynomials up to degree $N - 1$. For a Daubechies wavelet with N vanishing moments, this yields a system of $2N + 2$ equations in $2N$ unknowns (the filter has length $2N$), which is overdetermined but consistent. The solution for db2 ($N = 2$, yielding 4 coefficients) can be obtained analytically: $h = \frac{1}{4\sqrt{2}}[1 + \sqrt{3}, 3 + \sqrt{3}, 3 - \sqrt{3}, 1 - \sqrt{3}]^T$. For higher-order wavelets, the system is solved numerically, with different wavelet families distinguished by additional constraints: Daubechies wavelets select the minimum-phase solution (maximum energy concentration), Symlets minimize phase nonlinearity (near-symmetry), Coiflets impose vanishing moments on both the scaling function and wavelet

$$\sum_k k^m g[k] = 0 \quad (38)$$

and Biorthogonal wavelets relax orthogonality to

$$\sum_k h[k] \tilde{h}[k - 2m] = \delta[m] \quad (39)$$

$\tilde{h}[k]$ refers to synthesis kernel's coefficients. Note that since we have relaxed the orthogonality constraint the synthesis and analysis filter coefficients are now completely different. to achieve exact symmetry ($h[k] = h[N - 1 - k]$).

4.4 Implementation Pipeline (Theoretical Overview)

The proposed pipeline integrates DSP and Machine Learning concepts as follows:

1. **Preprocessing:** Normalization to standardize amplitude across varying EEG recordings.
2. **Denoising:** Application of the **VisuShrink** algorithm to remove high-frequency Gaussian noise while preserving sharp signal features.
3. **Decomposition:** Multi-level DWT to isolate specific frequency bands (0 – 60+ Hz).
4. **Feature Extraction:** Calculation of statistical metrics (Entropy, Energy) from the wavelet coefficients of each band.
5. **Classification:** Training supervised learning models (SVM, XGBoost) on the extracted feature vectors.

5 Implementation and Simulation Results

5.1 Dataset

The data used in this study is the **Epileptic Seizure Recognition Dataset** obtained from the <https://archive.ics.uci.edu/datasets>. The original dataset consists of EEG recordings from 500 distinct subjects. For each subject, brain activity was recorded for 23.6 seconds, sampled into a time-series of 4097 data points ($f_s = 173.61$ Hz).

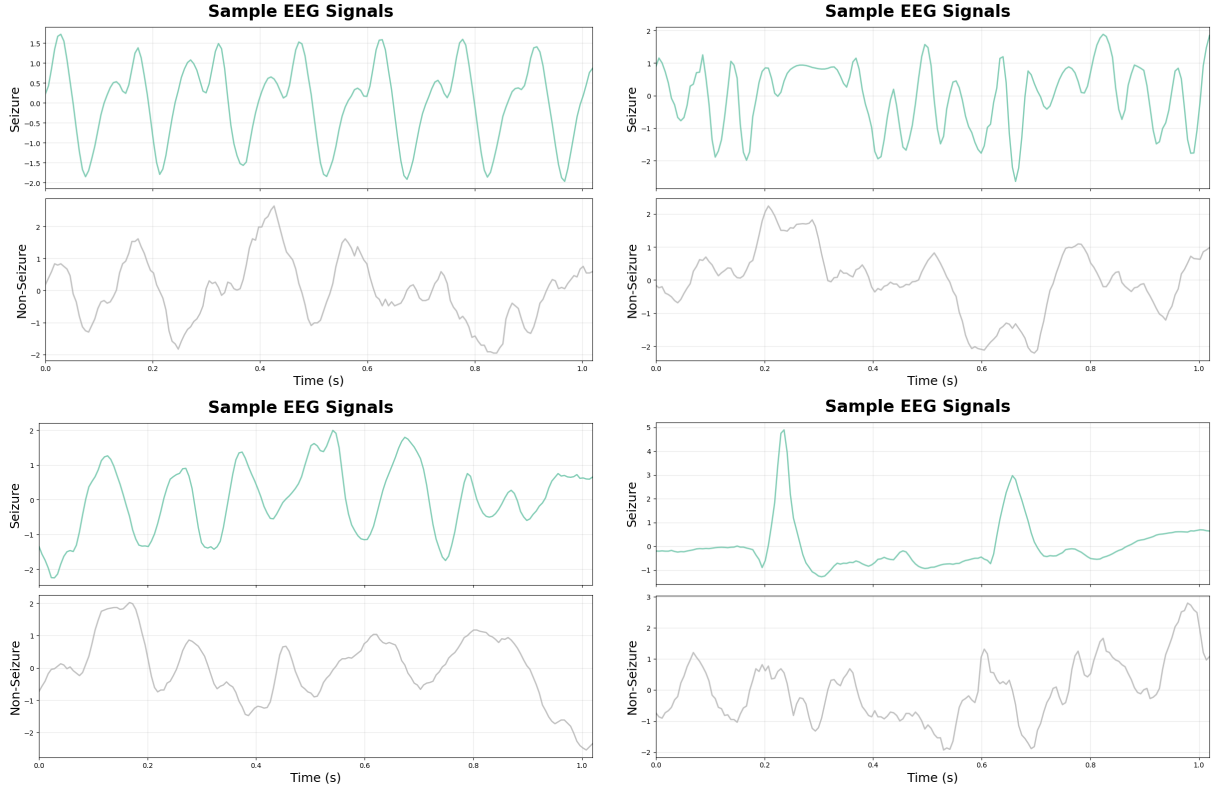


Figure 7: Comparison of Seizure (top row) and Non-Seizure (bottom row) EEG signals in time domain.

To facilitate machine learning analysis, the raw data underwent a restructuring process:

- **Segmentation:** The 4097 data points for each of the 500 subjects were divided into 23 non-overlapping chunks.
- **Chunk Dimensions:** Each resulting chunk contains 178 data points, representing approximately 1 second of EEG activity:

$$T = \frac{N}{f_s} = \frac{178}{173.61} \approx 1.025 \text{ seconds} \quad (40)$$

- **Total Volume:** This segmentation yielded a total matrix of 11,500 samples (rows) and 179 columns (178 explanatory variables $X_1 \dots X_{178}$ and 1 response variable y).

The response variable y contains the category labels $\{1, 2, 3, 4, 5\}$, defined as follows:

- **Class 5 (Eyes Open):** EEG recorded from healthy volunteers with eyes open.
- **Class 4 (Eyes Closed):** EEG recorded from healthy volunteers with eyes closed.
- **Class 3 (Healthy Region):** EEG recorded from the healthy brain hemisphere of patients with a tumor.
- **Class 2 (Tumor Region):** EEG recorded directly from the tumor site.
- **Class 1 (Seizure):** EEG recorded during active epileptic seizure activity.

5.2 Pipeline

5.2.1 Preprocessing

Although the dataset presents a multi-class problem, most existing literature focuses on the crucial distinction between seizure and non-seizure states. We mapped the classes to a binary problem as follows [26]:

$$y_{binary} = \begin{cases} 1(\text{Positive}) & \text{if } y_{original} = 1 \\ 0(\text{Negative}) & \text{if } y_{original} \in \{2, 3, 4, 5\} \end{cases} \quad (41)$$

This results in an imbalanced dataset with 2,300 seizure samples (20%) and 9,200 non-seizure samples (80%)[cite: 9]. To standardize the input features, we applied Z-score normalization to each 178-point signal vector $\mathbf{x}$ independently:

$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \mu_x}{\sigma_x} \quad (42)$$

5.2.2 Signal Denoising using VisuShrink

EEG signals are inherently non-stationary and contaminated by wide-band noise such as electromyograms (EMG) and ambient electrical interference[27]. To enhance the signal-to-noise ratio (SNR) while preserving sharp features like epileptic spikes, we implemented the **VisuShrink** method[28].

5.2.2.1 Mathematical Formulation

We model the observed discrete signal $y[n]$ as the sum of the true deterministic signal $f[n]$ and additive Gaussian white noise $z[n]$:

$$y[n] = f[n] + \sigma z[n], \quad z[n] \stackrel{i.i.d}{\sim} \mathcal{N}(0, 1) \quad (43)$$

Because the Discrete Wavelet Transform (DWT) is an orthogonal linear operator, this additive noise structure is preserved in the wavelet domain. The wavelet coefficients $w_{j,k}$ at level j and position k can be expressed as:

$$w_{j,k} = \theta_{j,k} + \sigma \epsilon_{j,k}, \quad \epsilon_{j,k} \sim \mathcal{N}(0, 1) \quad (44)$$

The core premise of wavelet denoising is **sparsity**: the energy of the true signal $f[n]$ is concentrated in a small number of large coefficients $\theta_{j,k}$, whereas the noise energy $\sigma \epsilon_{j,k}$ is spread uniformly across all coefficients with small amplitude.

5.2.2.2 Step 1: Robust Noise Estimation

To perform denoising, we must first estimate the noise level σ . Standard estimators (like the sample standard deviation) are unsuitable because they are sensitive to the large coefficients of the signal itself (outliers).

Instead, we use the **Median Absolute Deviation (MAD)** of the coefficients at the finest resolution level (cD_1) [29]. The first detail level consists primarily of high-frequency noise, making it a reliable estimator for the noise floor. The estimator is defined as:

$$\hat{\sigma} = \frac{\text{median}(|cD_1|)}{0.6745} \quad (45)$$

Intuition for MAD and cD_1 :

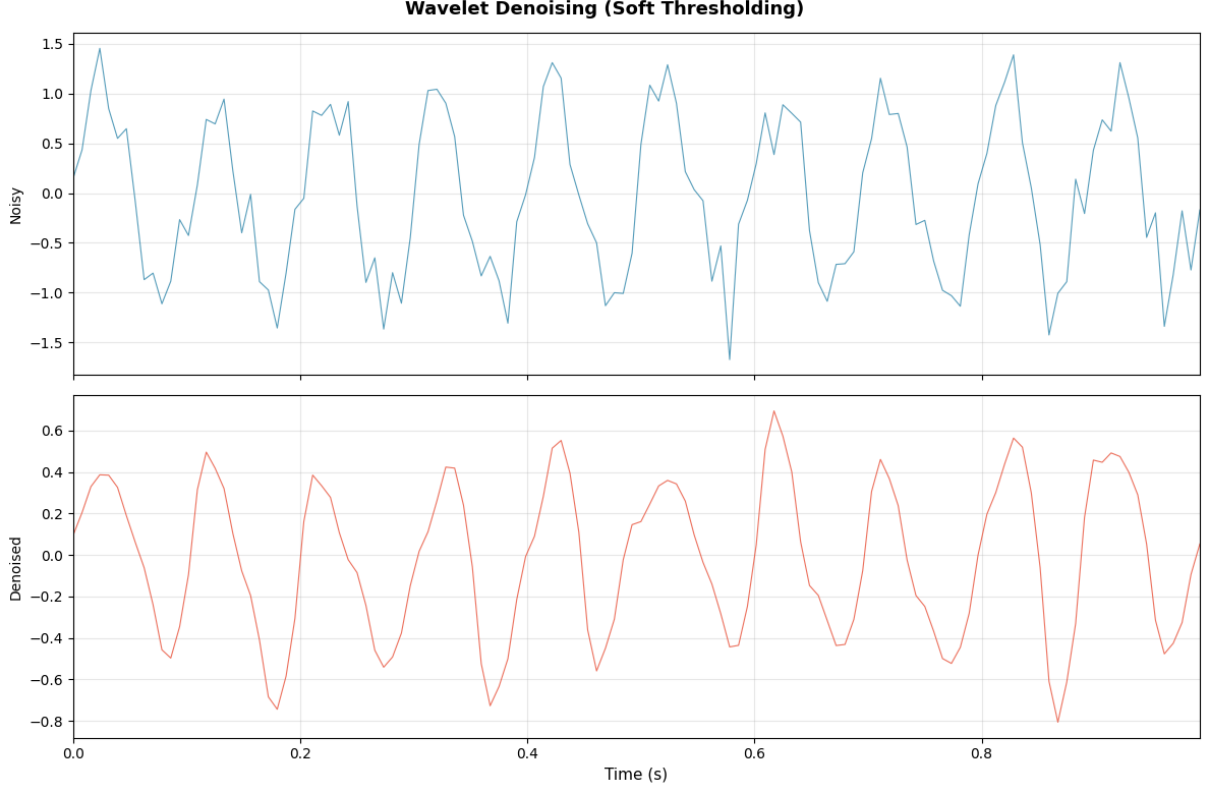


Figure 8: Effect of denoising with soft thresholding.

- **High-Frequency dominance:** High-frequency noise dominates the cD_1 detail level, while the biological EEG signal is minimal at these frequencies [30].
- **Robustness:** The median has a high breakdown point (50%), meaning it ignores the few large "signal" coefficients (ictal spikes) and focuses on the "bulk" of the noise distribution.
- **Normalizing Factor:** The factor 0.6745 is the value of $\Phi^{-1}(0.75)$, ensuring that $\hat{\sigma}$ is a consistent estimator of the standard deviation σ for Gaussian noise.

Therefore, $\text{median}(|Z|)/0.6745 \approx 1$, providing a consistent estimate of σ [cite: 36].

5.2.2.3 Step 2: Universal Threshold (VisuShrink)

We utilize the Universal Threshold λ , derived from Extreme Value Theory. For a sequence of N independent Gaussian variables, the probability that the maximum amplitude of pure noise exceeds $\sqrt{2 \ln N}$ tends to zero asymptotically [28]:

$$\lim_{N \rightarrow \infty} P \left(\max_{1 \leq i \leq N} |z_i| > \sqrt{2 \ln N} \right) = 0 \quad (46)$$

This allows us to define a probabilistic "ceiling" for the noise:

$$\lambda = \hat{\sigma} \sqrt{2 \ln(N)} \quad (47)$$

With $N = 178$, this threshold ensures that with high probability, any coefficient $|w_{j,k}| < \lambda$ is attributable to noise and should be removed.

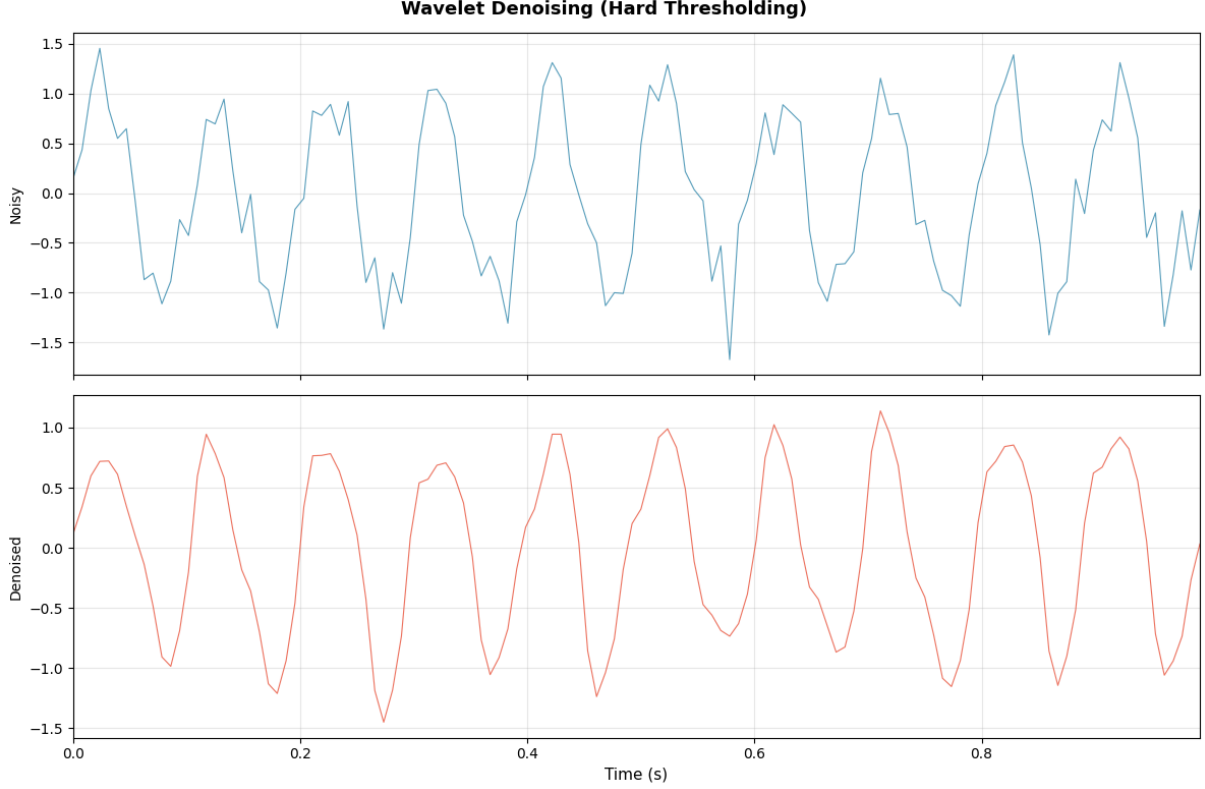


Figure 9: Effect of denoising with hard thresholding.

5.2.2.4 Step 3: Soft Thresholding

To reconstruct the signal, we apply the Soft Thresholding operator η_S to the detail coefficients cD_j . Unlike hard thresholding, which creates discontinuities (Gibbs oscillations) by abruptly chopping coefficients, soft thresholding shrinks them towards zero, preserving the smoothness of the reconstructed signal 8:

$$\eta_S(w, \lambda) = \text{sgn}(w) \cdot \max(|w| - \lambda, 0) \quad (48)$$

The approximation coefficients cA_J are left unthresholded to preserve the low-frequency baseline of the EEG signal [27]. The denoised signal $\hat{f}$ is obtained via the Inverse Discrete Wavelet Transform (IDWT) of the processed coefficients. Note that hard thresholding with which all the coefficients below the threshold are zeroed out was also tried. We opted not to use it in the final pipeline since it over-smoothed the signal ignoring its content 9.

5.2.3 Frequency Band Separation and EEG Brain Waves

Electroencephalogram (EEG) signals are non-stationary time-series data composed of distinct oscillatory patterns associated with specific physiological and cognitive states [31]. The Discrete Wavelet Transform (DWT) is particularly well-suited for EEG analysis because its multi-resolution capability naturally separates these components into frequency bands corresponding to standard clinical definitions: Delta (δ), Theta (θ), Alpha (α), Beta (β), and Gamma (γ) [27].

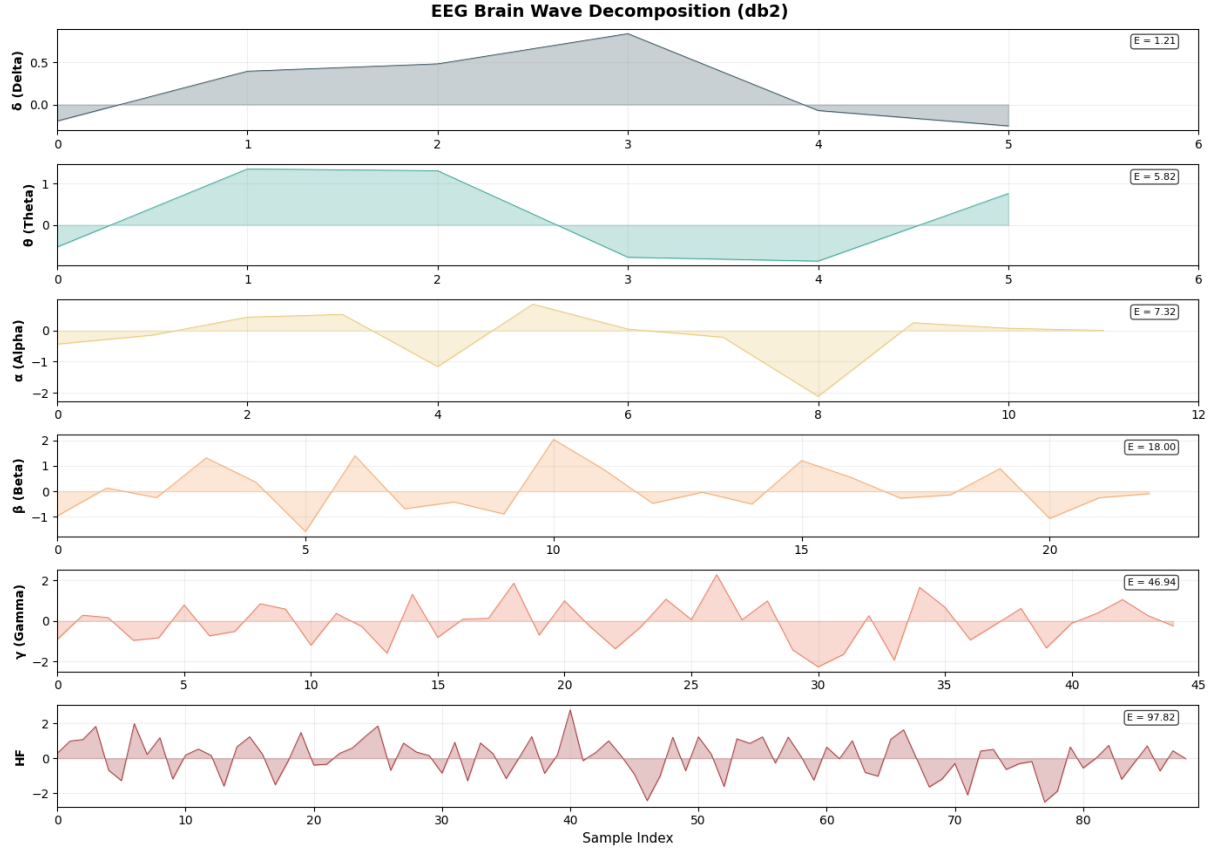


Figure 10: Approximation and detail coefficients of DWT and corresponding bands. Note that number of coefficients halves in each level due to downsampling in Mallat’s algorithm.

5.2.3.1 DWT Level to Frequency Mapping

The frequency range covered by the detail coefficients at decomposition level j is determined by the sampling frequency f_s according to the dyadic relationship:

$$f_{range}(j) = \left[\frac{f_s}{2^{j+1}}, \frac{f_s}{2^j} \right] \quad (49)$$

Given our sampling rate of $f_s = 173.61$ Hz, a 5-level decomposition using the Daubechies 4 (db4) wavelet yields the following frequency mapping:

Table 1: DWT Decomposition Levels and Corresponding EEG Bands

Level	Coeffs	Freq. Range (Hz)	Mapped Brain Wave
Level 1	cD_1	43.4 – 86.8	HF / Noise
Level 2	cD_2	21.7 – 43.4	Gamma (γ)
Level 3	cD_3	10.8 – 21.7	Beta (β)
Level 4	cD_4	5.4 – 10.8	Alpha (α)
Level 5	cD_5	2.7 – 5.4	Theta (θ)
Approx.	cA_5	0 – 2.7	Delta (δ)

Note on Decomposition Depth: The maximum possible decomposition level J_{max} is constrained by the length of the signal N and the length of the wavelet filter L . Specifi-

cally, the decomposition must stop when the number of coefficients becomes smaller than the filter length:

$$J_{max} = \lfloor \log_2 \left(\frac{N}{L-1} \right) \rfloor \quad (50)$$

For our signal length of $N = 178$, using a longer filter (e.g., Coiflet 4) would result in fewer available decomposition levels compared to a shorter filter like Haar or db4, potentially merging the lower frequency bands (Delta/Theta).

5.2.3.2 Physiological Relevance for Seizure Detection

Decomposing the signal into these bands allows us to isolate seizure-specific biomarkers [32]:

- **Ictal Discharges:** Seizures are often characterized by high-amplitude rhythmic discharges in the Theta and Alpha bands (cD_4, cD_5).
- **Post-Ictal Slowing:** Following a seizure, brain activity often slows down, resulting in increased energy in the Delta band (cA_5).
- **High-Frequency Oscillations (HFOs):** Bursts of energy in the Gamma band (cD_1) are critical indicators of the seizure onset zone.

5.2.4 Feature Extraction

Following denoising and decomposition, we extract statistical features from the wavelet coefficients of each sub-band. While the raw coefficients capture the time-frequency information, they are high-dimensional and unsuitable for direct classification. Statistical features compress this information into a compact vector that characterizes the signal's behavior [31].

5.2.4.1 Feature Selection Strategy

We initially explored an extensive feature set including skewness, kurtosis, mean, and root mean square (RMS). However, preliminary experiments indicated that expanding the feature space beyond energy and entropy measures yielded diminishing returns in classification accuracy while increasing computational complexity. Consequently, we selected a robust 3-feature vector per sub-band:

1. **Normalized Energy (E_{rel}):** Epileptic seizures involve a massive synchronization of neurons, causing a redistribution of signal energy across frequency bands. We calculate the relative energy of each band to capture this spectral shift:

$$E_{rel}^{(j)} = \frac{\sum_k |w_{j,k}|^2}{E_{total}}, \quad \text{where } E_{total} = \sum_j \sum_k |w_{j,k}|^2 \quad (51)$$

Using relative energy makes the feature robust to variations in absolute signal amplitude between different patients.

2. **Shannon Entropy (H):** Entropy quantifies the complexity or "randomness" of the signal distribution within a band. Seizure states typically exhibit lower entropy

in specific bands due to the highly rhythmic and ordered nature of ictal discharges compared to the chaotic background activity of a healthy brain [32].

$$H^{(j)} = - \sum_k p_k \log_2(p_k), \quad \text{where } p_k = \frac{|w_{j,k}|^2}{\sum_m |w_{j,m}|^2} \quad (52)$$

Here, p_k represents the probability distribution of energy within the j -th sub-band.

3. **Standard Deviation (σ):** This metric captures the variability of the coefficients. High-amplitude seizure spikes significantly increase the standard deviation in the affected frequency bands.

$$\sigma^{(j)} = \sqrt{\frac{1}{M} \sum_{k=1}^M (w_{j,k} - \mu_j)^2} \quad (53)$$

Where M is the number of coefficients in sub-band j and μ_j is the mean of the coefficients (typically close to zero for detail levels).

5.2.4.2 Feature Vector Construction

For a standard 5-level decomposition, we obtain 6 sets of coefficients ($cD_1, cD_2, cD_3, cD_4, cD_5, cA_5$). Extracting 3 features from each yields an 18-dimensional feature vector per EEG sample:

$$\mathbf{F} = [E_{D1}, H_{D1}, \sigma_{D1}, \dots, E_{A5}, H_{A5}, \sigma_{A5}]^T \quad (54)$$

This dimensionality reduction (from 178 raw samples to 18 features) retains the most discriminative information while significantly reducing the risk of the "curse of dimensionality" during classifier training.

5.2.5 Classification Methods

We implemented three robust classification algorithms, each offering distinct advantages for handling high-dimensional EEG feature spaces.

5.2.6 Support Vector Machine (SVM)

SVM constructs an optimal hyperplane to maximize the margin between Seizure and Non-Seizure classes [33]. To handle non-linear decision boundaries, we employed the Radial Basis Function (RBF) kernel. The model was configured with a regularization parameter of $C = 1.0$ and a scaled gamma coefficient to normalize feature variance during the optimization process.

5.2.7 XGBoost (Extreme Gradient Boosting)

XGBoost is an ensemble method that builds decision trees sequentially, where each new tree attempts to correct the residual errors of the previous ensemble [34]. We configured the model with 100 estimators, a maximum tree depth of 6, and a learning rate of 0.1. This setup allows the model to capture complex non-linear interactions by minimizing a regularized objective function that penalizes model complexity.

5.2.8 Random Forest

Random Forest constructs an ensemble of independent decision trees trained on bootstrap samples of the data [35]. Unlike boosting, trees are grown in parallel using random feature subsets at each split node. We utilized 100 estimators and aggregated predictions via majority voting, a strategy that effectively reduces variance and ensures robustness against the noise and outliers typical in EEG signals.

5.2.9 Validation Strategy

5.2.9.1 5-Fold Stratified Cross-Validation

Given the class imbalance (20% Seizure vs. 80% Non-Seizure), we utilized **Stratified 5-Fold Cross-Validation** [36]. This ensures that each fold maintains the original class distribution, preventing bias in the performance estimates.

The procedure is as follows:

1. The dataset is split into 5 disjoint subsets (folds).
2. In each iteration k (from 1 to 5), the model is trained on 4 folds and validated on the remaining k -th fold.
3. Performance metrics are averaged across all 5 folds to report the final Mean $\pm$ Standard Deviation.

5.2.9.2 Performance Metrics

We evaluated the models using five standard metrics to ensure clinical reliability:

- **Accuracy:** The overall proportion of correct predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (55)$$

- **Precision:** The probability that a predicted seizure is actually a seizure (minimizing false alarms).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (56)$$

- **Recall (Sensitivity):** The probability of detecting an actual seizure (minimizing missed diagnoses).

$$\text{Recall} = \frac{TP}{TP + FN} \quad (57)$$

- **F1 Score:** The harmonic mean of Precision and Recall, providing a balanced metric for imbalanced datasets.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (58)$$

- **ROC-AUC:** The Area Under the Receiver Operating Characteristic Curve, calculated as the integral of the True Positive Rate (TPR) against the False Positive Rate (FPR):

$$AUC = \int_0^1 TPR(FPR) dFPR \quad (59)$$

5.3 Results and Analysis

5.3.1 CWT Scalogram Analysis

Using the Morlet wavelet, we generated Scalograms ($|W(a, b)|^2$) to visualize the energy distribution. As hypothesized in the methodology, the seizure signals exhibit distinct high-energy “hotspots” across scales, contrasting with the diffuse energy of healthy signals.

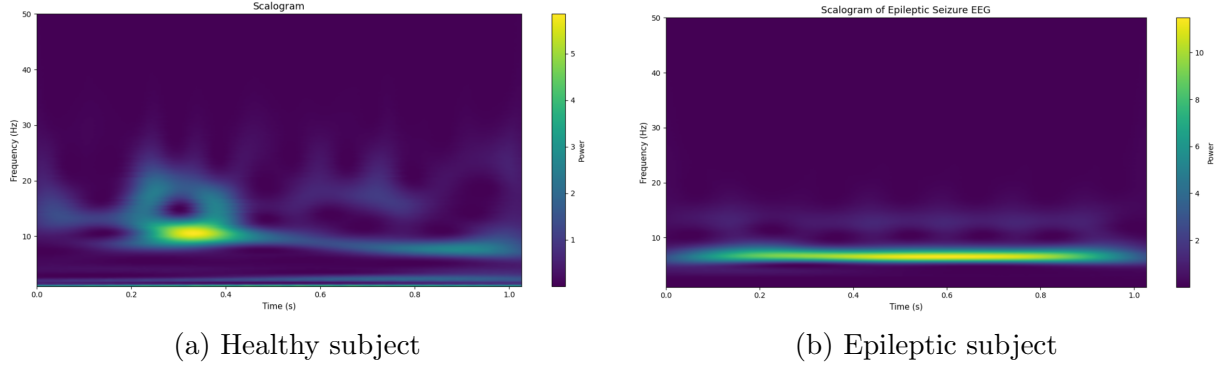


Figure 11: **Scalogram Comparison.** (a) Scalogram of a Healthy subject showing low-energy, sporadic background activity with occasional bursts. (b) Scalogram of an Epileptic subject showing intense, sustained energy bursts in the 5-20 Hz range.

5.3.2 Dimensionality Reduction & Visualization

To validate the discriminative power of our feature set prior to classification, we projected the high-dimensional feature space ($\mathbb{R}^{18}$) into 2D space using both linear and non-linear manifold learning techniques: **Principal Component Analysis (PCA)** and **t-Distributed Stochastic Neighbor Embedding (t-SNE)**.

For PCA, we performed an eigendecomposition of the covariance matrix C :

$$C = \frac{1}{n-1} X^T X = V \Lambda V^T \quad (60)$$

The projection onto the first two principal components accounts for approximately 58% of the total variance. As seen in Figure 12 (Left), PCA reveals a gradient of separation but exhibits significant overlap between classes, indicating that the decision boundary is likely non-linear.

In contrast, t-SNE minimizes the Kullback-Leibler divergence between the joint probabilities of the low-dimensional embedding and the high-dimensional data, effectively preserving local neighborhood structures. Figure 12 (Right) demonstrates that t-SNE successfully disentangles the data into two distinct, well-separated clusters. This topological separability confirms that while the classes may not be linearly separable (as suggested by PCA), the extracted wavelet features possess sufficient discriminative information for non-linear classifiers like XGBoost to establish accurate decision boundaries.

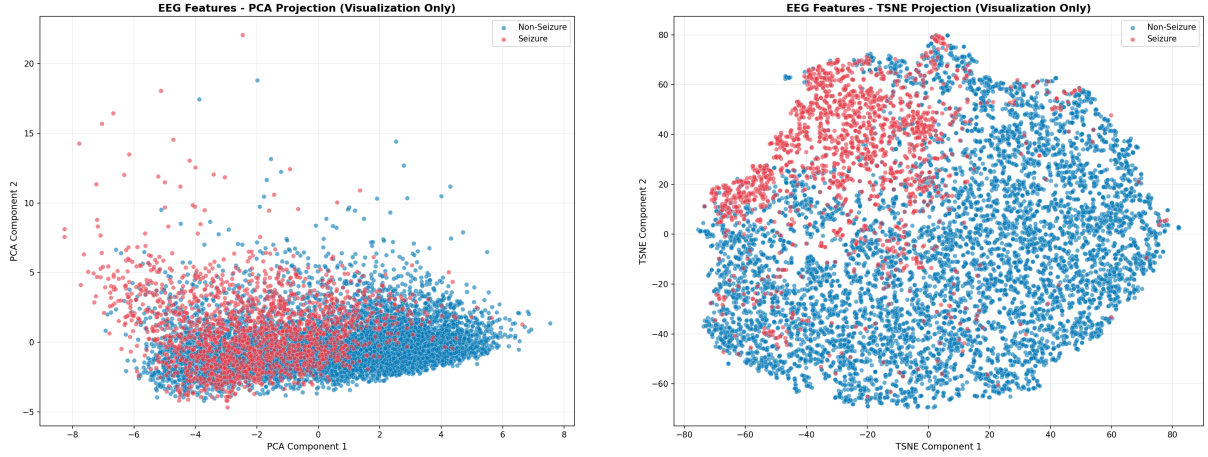


Figure 12: **2D Feature Projection comparison.** Left: PCA projection showing linear overlap between classes. Right: t-SNE projection revealing distinct, separable clusters for Seizure (Red) and Non-Seizure (Blue) classes, validating the non-linear separability of the wavelet features.

5.3.3 Classification Results

The proposed pipeline was evaluated using Stratified 5-Fold Cross-Validation. The performance metrics for the XGBoost classifier across different wavelet families are summarized in Table 2.

Table 2: XGBoost Cross-Validation Performance Comparison (Mean $\pm$ Std)

Wavelet	CV Accuracy	CV Precision	CV Recall	CV F1-Score	CV ROC AUC
db2	0.962 $\pm$ 0.003	0.928 $\pm$ 0.007	0.877 $\pm$ 0.013	0.902 $\pm$ 0.009	0.987 $\pm$ 0.002
sym2	0.960 $\pm$ 0.003	0.922 $\pm$ 0.004	0.874 $\pm$ 0.014	0.897 $\pm$ 0.008	0.987 $\pm$ 0.002
coif2	0.956 $\pm$ 0.004	0.906 $\pm$ 0.016	0.873 $\pm$ 0.010	0.889 $\pm$ 0.010	0.985 $\pm$ 0.002
haar	0.947 $\pm$ 0.005	0.897 $\pm$ 0.015	0.830 $\pm$ 0.013	0.862 $\pm$ 0.013	0.974 $\pm$ 0.005
bior1.1	0.947 $\pm$ 0.005	0.897 $\pm$ 0.015	0.830 $\pm$ 0.013	0.862 $\pm$ 0.013	0.974 $\pm$ 0.005

The db2 wavelet yielded the highest performance, achieving an F1-Score of 0.902 ± 0.009 . The confusion matrix for the best performing model (Figure 13) indicates a high specificity, with 98.70% of Non-Seizure segments correctly classified. The model identified 86.74% of seizure segments correctly, with a misclassification rate of 13.26%.

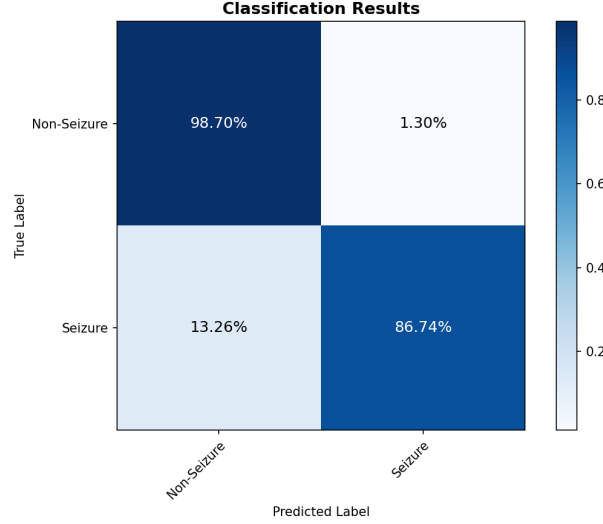


Figure 13: **Confusion Matrix (db2 + XGBoost)**. The model demonstrates high specificity (98.70%) in identifying healthy signals, which is critical for minimizing false alarms in clinical monitoring systems.

5.3.3.1 Impact of Wavelet Family and Length

We analyzed the impact of wavelet choice on classification performance. As illustrated in Figure 14, short-filter wavelets such as db2 and sym2 consistently outperform others. Notably, the Haar wavelet (equivalent to db1) performs significantly worse than other length-2 wavelets, likely due to its discontinuity, which fails to capture the smooth morphological features of neural oscillations.

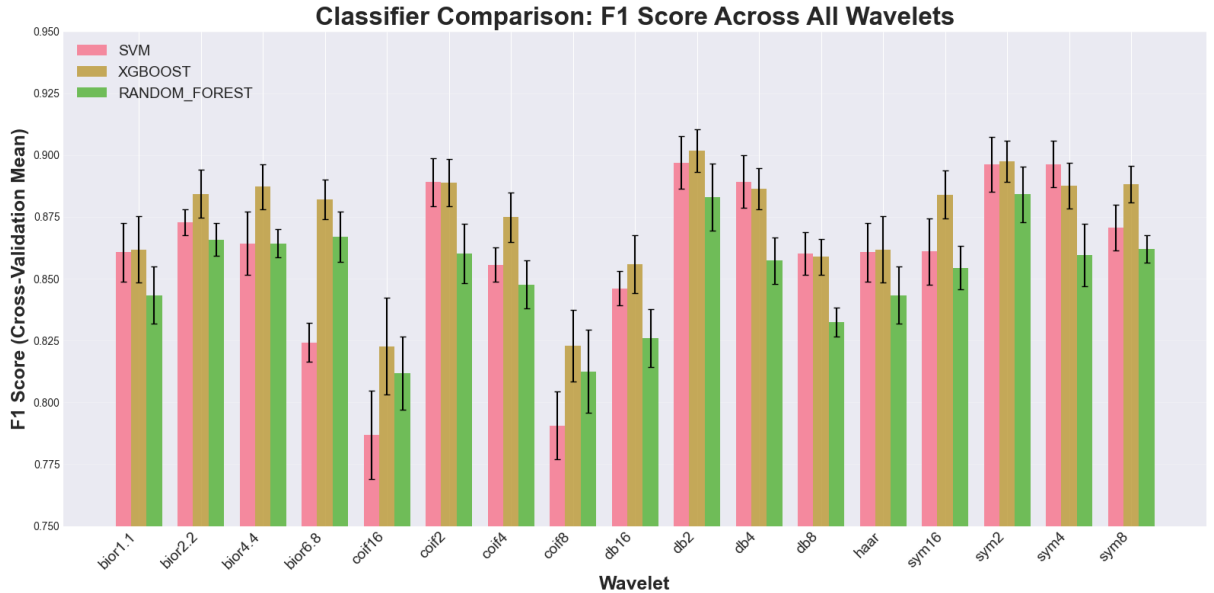


Figure 14: **Performance across Wavelet Families**. The Daubechies (db) and Symlet (sym) families generally outperform Coiflets and Biorthogonal wavelets, with db2 achieving the peak F1-score.

A distinct inverse relationship between filter length and classification performance is

observed in Figure 15. As the filter length increases (e.g., moving from **db2** to **db16**), accuracy and F1-score degrade.

This phenomenon can be attributed to the mathematical constraints of the Discrete Wavelet Transform on short signal segments. The maximum decomposition level J_{max} is defined as $\lfloor \log_2(N/(L-1)) \rfloor$. For our signal length of $N = 178$:

- A short filter like **db2** ($L = 4$) allows for 5 levels of decomposition, isolating the crucial Delta ($0 - 2.7$ Hz) and Theta ($2.7 - 5.4$ Hz) bands.
- A long filter like **db8** ($L = 16$) restricts decomposition to only 3 levels, effectively merging the Delta, Theta, and Alpha bands into a single approximation coefficient.

Since epileptic seizures are often characterized by rhythmic activity in the Theta range and post-ictal slowing in the Delta range, the inability to resolve these low-frequency bands with long filters results in a loss of critical diagnostic information.

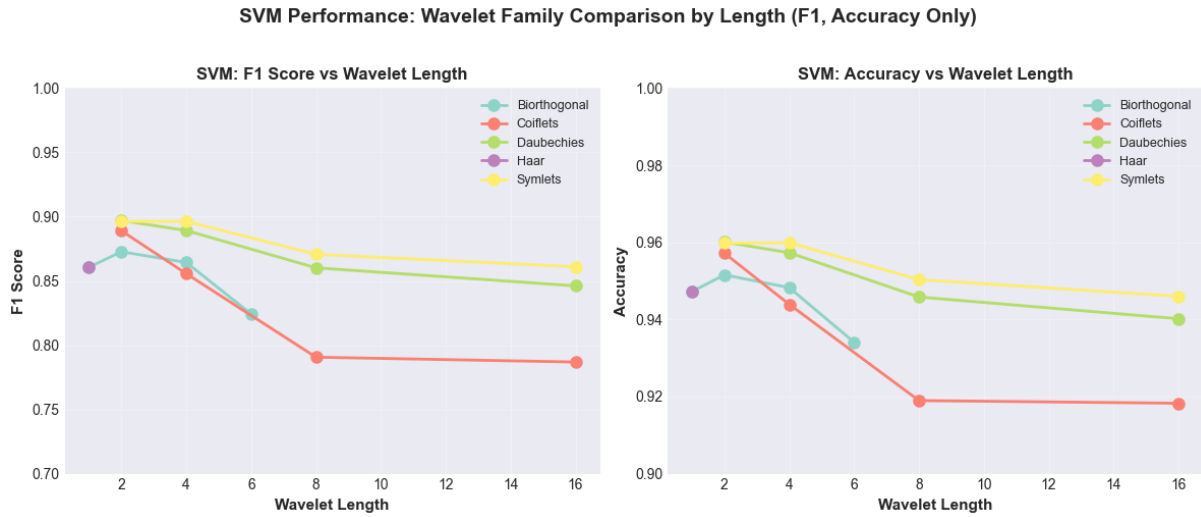


Figure 15: Effect of Filter Length on Performance. There is a clear downward trend in F1-Score as wavelet filter length increases. Longer filters reduce the maximum possible decomposition depth for short signals ($N = 178$), leading to poor frequency resolution in the biologically significant Delta and Theta bands.

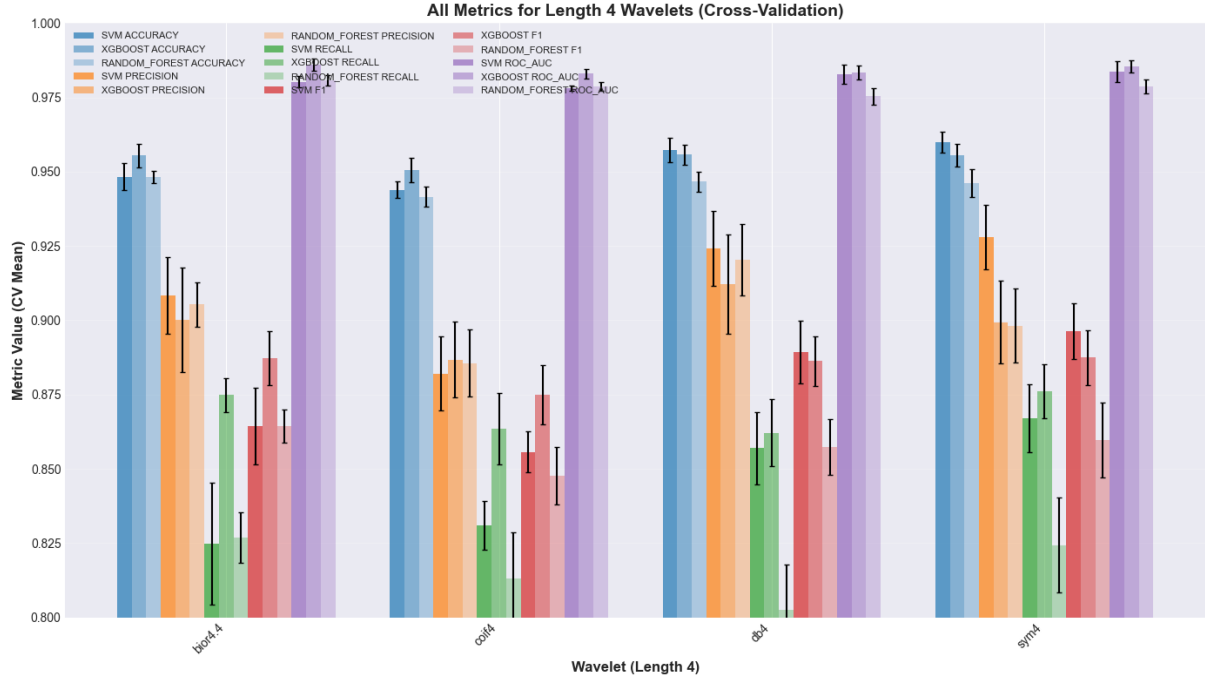


Figure 16: **Comprehensive Metric Comparison for Length-4 Wavelets.** Consistency is observed across all metrics (Accuracy, Precision, Recall, AUC) for length-4 wavelets, with db2 showing a slight advantage in Recall.

5.3.4 Most Relevant Features

In order to analyze which features are more important we have looked at the loading of each feature which is the sum of projections into principal components weighted by the explained variance of that principal component.

We observe that lower detail coefficient contribute slightly more than higher scales which makes one tend to deduce high frequency components are more discriminative¹⁷. However, there is no significant difference between contributions and it would be incorrect to draw a conclusion. All the coefficients contribute similarly and all of them are required for a true assessment.

5.3.5 Comparison with Fourier Transform

To verify the necessity of time-frequency analysis, we benchmarked our wavelet pipeline against a traditional Fourier Transform (FT) approach. Features were extracted from the same frequency bands using the Fast Fourier Transform (FFT) without temporal localization.

As shown in Figure 18, the FT-based model failed to classify seizures effectively, yielding a seizure detection accuracy of only 51.52%—equivalent to random guessing. This stark contrast empirically validates the theoretical assertion that EEG signals are non-stationary. The transient nature of epileptic spikes requires the temporal localization provided by the Wavelet Transform, which the Fourier Transform inherently lacks.

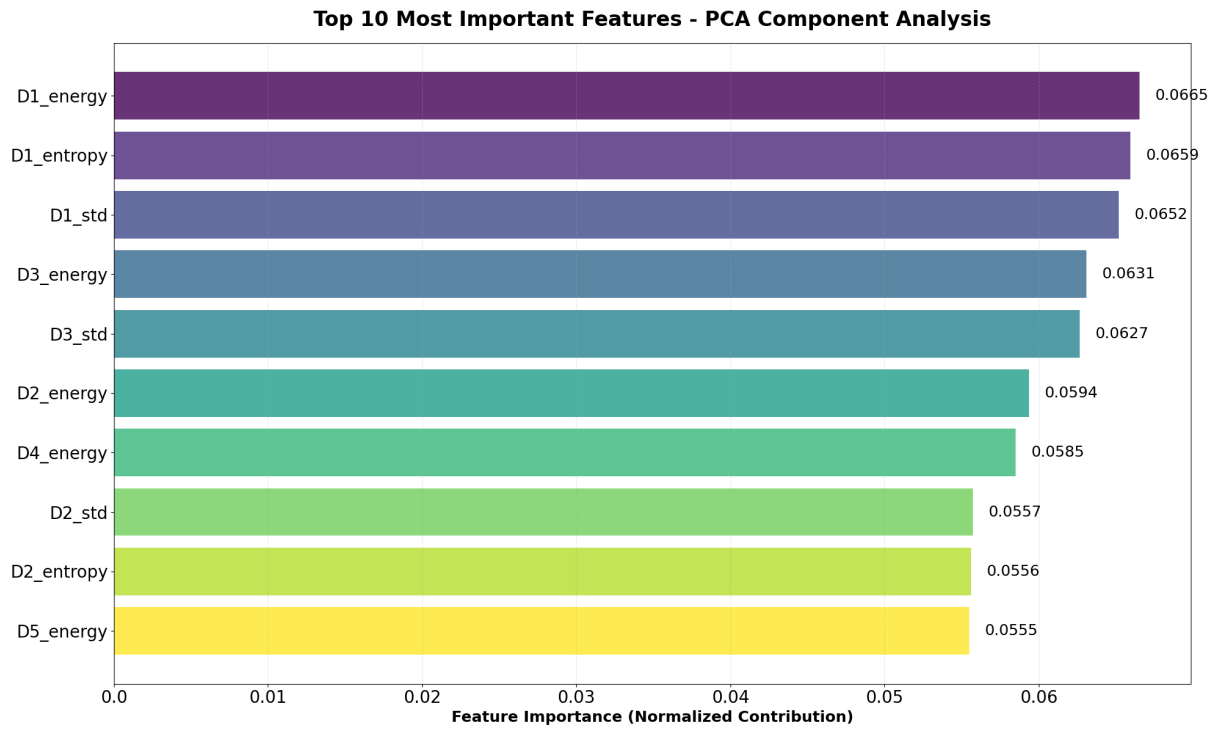


Figure 17: Importance of features

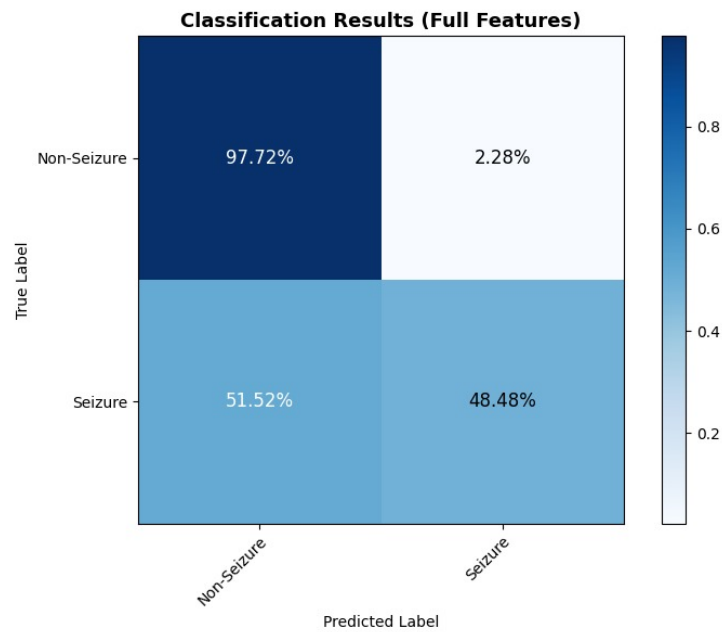


Figure 18: **Confusion Matrix (Fourier Transform Features)**. The poor sensitivity (51.52%) highlights that frequency information alone is insufficient for detecting transient seizure events, confirming the superiority of the Wavelet Transform for non-stationary EEG analysis.

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7 Code

Code is submitted seperately and also can be found in [here](#)